

Chapter 10: Next-Generation IoMT Solutions for Real-Time Diabetes Monitoring and Decision Support

Sathiyamurthi K¹, Venkatesan R^{1*}, Sharmi B¹, Ithaya Priya B²

¹Department of Biomedical Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India.

²Department of Computer Science and Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India.

*Corresponding Author mail : vengatesan.r@egspec.org

Abstract

The increasing prevalence of chronic diseases, particularly diabetes mellitus, has created a need for intelligent and scalable healthcare systems. The Internet of Medical Things (IoMT) enables continuous patient monitoring, real-time data collection, and improved clinical decision-making. This chapter presents a systematic review of IoMT applications for diabetes management using a multi-method approach that combines structural coding, Quanteda-based text mining, Latent Dirichlet Allocation (LDA), and the Fuzzy Analytic Hierarchy Process (FAHP). The analysis shows that wearable sensors are the core components of IoMT systems, while cloud, fog, and edge computing support efficient data processing and timely healthcare services. LDA identified major research themes related to patient monitoring and data-driven healthcare. The study highlights the importance of adopting edge-enabled IoMT architectures to improve response time, data privacy, and system efficiency. Overall, this chapter provides a clear overview of current IoMT technologies and future research directions for developing reliable, scalable, and patient-centered healthcare systems.

Keywords: Internet of Medical Things (IoMT), Wearable Sensors, Edge Computing, Latent Dirichlet Allocation (LDA), Fuzzy Analytic Hierarchy Process (FAHP).

1. Introduction

The Internet of Things (IoT) has become an important technology in modern healthcare by connecting wearable sensors, medical devices, and mobile applications. This connectivity supports continuous health monitoring, real-time data collection, and communication between patients and healthcare providers, leading to improved disease management and personalized healthcare services [1], [2].

IoT is especially valuable in the management of chronic diseases, where regular monitoring and timely medical intervention are essential. By collecting and analysing patient data, IoT systems help healthcare professionals make informed decisions, improve treatment outcomes, and enhance the quality of life of patients [3]–[5].

The growing number of chronic diseases, particularly in developing countries, has increased the demand for efficient and affordable healthcare solutions[6]. IoT-based healthcare systems offer a practical solution by enabling remote monitoring, reducing healthcare costs, and supporting early diagnosis and timely treatment [7], [8].

This chapter provides a systematic review of IoT applications in chronic disease management, with a focus on diabetes mellitus. It discusses current technologies, research trends, and the role of edge intelligence in improving real-time decision-making, reducing latency, and enhancing data privacy in healthcare systems.

2. Key Applications of IoMT in Healthcare

2.1 Remote Patient Monitoring

IoMT enables continuous monitoring of patients through wearable sensors and connected medical devices that collect real-time health data, such as vital signs and disease-related parameters. Continuous monitoring is particularly beneficial for patients with chronic diseases, improving long-term disease management and preventive care [9], [10].

2.2 Telemedicine and Telehealth

IoMT supports telemedicine by enabling remote consultations, real-time patient monitoring, and secure communication between patients and healthcare providers. This improves access to healthcare services, especially in rural and underserved areas, while reducing travel costs and increasing patient convenience [11].

2.3 Chronic Disease Management

IoMT plays an important role in managing chronic diseases such as diabetes, cardiovascular disorders, and respiratory illnesses. Wearable and implantable devices continuously collect patient data, enabling early detection of health abnormalities, personalized treatment, and long-term disease monitoring. These capabilities improve disease control and enhance patient quality of life.

2.4 Medication Management

Smart medication systems, including connected pill dispensers and mobile health applications, help patients follow prescribed treatment schedules by providing reminders and monitoring medication usage. These systems reduce medication errors, improve treatment adherence, and allow healthcare providers to monitor patient compliance remotely, resulting in better clinical outcomes [12], [13].

2.5 Data Analytics and Predictive Intelligence

IoMT devices generate large volumes of healthcare data that can be analysed using machine learning and predictive analytics. These techniques support early disease detection, monitor disease progression, and assist clinical decision-making. The integration of edge computing further improves system performance by reducing latency, minimizing cloud dependence, and enhancing data privacy [14].

2.6 Operational Efficiency

IoMT improves healthcare operations by automating routine tasks, optimizing resource utilization, and simplifying clinical workflows. These improvements reduce operational costs, increase efficiency, and allow healthcare professionals to devote more time to patient care, leading to better healthcare services and outcomes [15].

3. IoMT Architecture for Diabetes Management

The Internet of Medical Things (IoMT) architecture supports continuous glucose monitoring, real-time healthcare, and intelligent clinical decision-making for diabetes management. It consists of four main layers: the sensing layer, edge/fog layer, cloud and AI layer, and communication layer. Together, these layers enable efficient data collection, processing, storage, and transmission while improving system performance and data security [16], [17].

3.1 Sensing Layer

The sensing layer collects real-time physiological data using wearable and implantable medical devices. In diabetes care, Continuous Glucose Monitoring (CGM) systems, smart insulin pumps, non-invasive glucose sensors, and wearable fitness trackers continuously measure glucose levels and other health parameters [18]. Reliable data collection at this layer is essential for accurate monitoring and clinical decision-making.

3.2 Edge/Fog Layer

The edge/fog layer processes data close to the patient using smartphones, gateways, or embedded devices, reducing the need to transmit all data to the cloud [19]. It performs preliminary analysis, detects abnormal glucose levels, generates immediate alerts, and filters data before transmission. Local processing reduces latency and enables faster clinical responses.

3.3 Cloud and AI Layer

The cloud layer stores long-term patient records and supports advanced data analysis using artificial intelligence (AI) and machine learning (ML) techniques [20]. It enables healthcare providers to monitor patients remotely, predict glucose trends, generate personalized treatment recommendations, and integrate patient information with electronic health records (EHR).

3.4 Communication Technologies

Wireless communication enables reliable data exchange between IoMT devices and healthcare systems [21]. Bluetooth Low Energy (BLE) is widely used for wearable devices because of its low power consumption. The choice of communication technology depends on the required transmission range, energy efficiency, and deployment environment.

The integration of these layers creates an efficient IoMT system for diabetes management. By combining wearable sensing, edge computing, cloud analytics, and reliable communication technologies, IoMT enables timely decision-making, improves patient monitoring, enhances data privacy, and supports effective healthcare delivery in both urban and rural settings [22].

4. Methodology

This study employs a multi-stage methodology to analyse the use of Internet of Things (IoT) technologies in chronic disease management. Both qualitative and quantitative methods were used, including structural coding, Quanteda-based text mining, Latent Dirichlet Allocation (LDA), and the Fuzzy Analytic Hierarchy Process (FAHP). These techniques were applied to identify research trends, classify IoT applications, and prioritise key technologies for chronic disease management [23].

4.1 Structural Coding Approach

A systematic literature review was conducted to examine recent developments in IoT-based chronic disease management. Since wireless sensor networks are a core component of IoT, both search terms were included to improve the coverage of relevant studies [24].

A total of 161 relevant studies were selected and classified according to the World Health Organization (WHO) list of chronic diseases. Additional searches were performed through Google

Scholar when literature on specific diseases was limited. The overall workflow of the structural coding process is illustrated in Fig. 1. Structural coding was then applied to identify recurring technologies, research patterns, and disease-specific applications, enabling consistent extraction of relevant information while reducing ambiguity in technology identification.

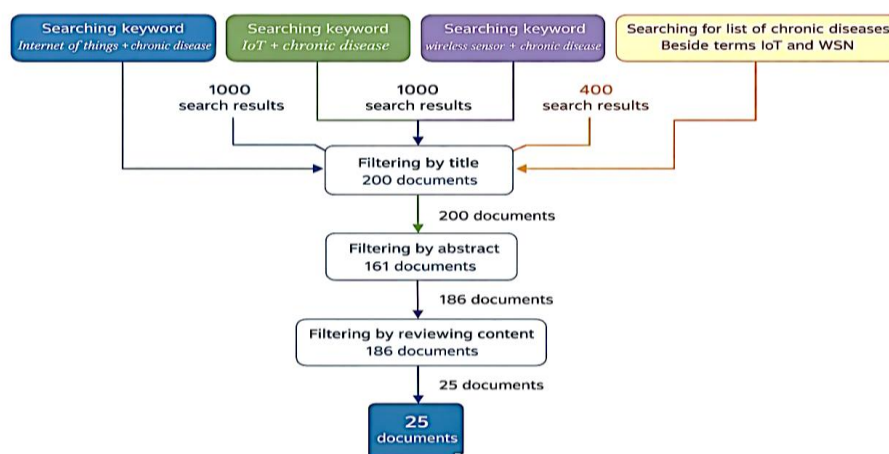


Fig. 1. Structural coding framework for systematic identification of IoMT technologies in chronic disease literature.

4.2 Quanteda-Based Text Mining

The selected publications were converted into plain text and analysed using the R programming environment with the Quanteda package. This approach enabled efficient processing of large text datasets while improving the identification of disease-specific IoMT applications. Manual verification was performed to minimise errors caused by incidental disease mentions, ensuring reliable classification of the literature [26], [27].

4.3 Latent Dirichlet Allocation (LDA) for Topic Modelling

Latent Dirichlet Allocation (LDA) was used to identify the major research themes in IoMT-based chronic disease management. The optimal number of topics was determined using coherence and perplexity measures, while word clouds were generated to visualise frequently occurring terms as given in Fig.2.

The LDA analysis identified important research areas, including remote patient monitoring, wearable healthcare devices, data security, privacy, and disease-specific IoMT applications. The extracted topics provided a structured understanding of current research trends and highlighted future opportunities for developing intelligent IoMT healthcare systems [19], [21].

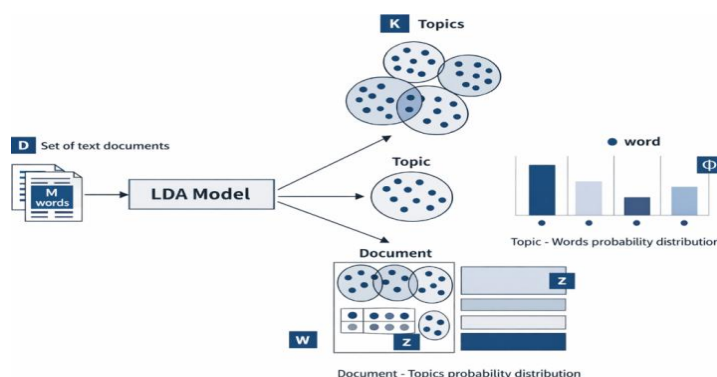


Fig. 2. Workflow of Latent Dirichlet Allocation (LDA) for topic extraction from IoMT literature.

4.4 Fuzzy Analytic Hierarchy Process (FAHP) for Prioritizing Chronic Diseases in IoT Applications:

The Fuzzy Analytic Hierarchy Process (FAHP) was used to prioritise chronic diseases based on their suitability for IoMT applications. FAHP combines the Analytic Hierarchy Process (AHP) with fuzzy logic to handle uncertainty in expert judgments, providing more reliable decision-making [16]. The FAHP workflow used in this study is shown in Fig. 3.

The FAHP analysis ranked the chronic diseases according to their priority for IoMT implementation. Among the evaluated diseases, cardiovascular diseases received the highest priority, indicating their significant healthcare burden and the potential benefits of IoMT technologies in improving monitoring and disease management [25].

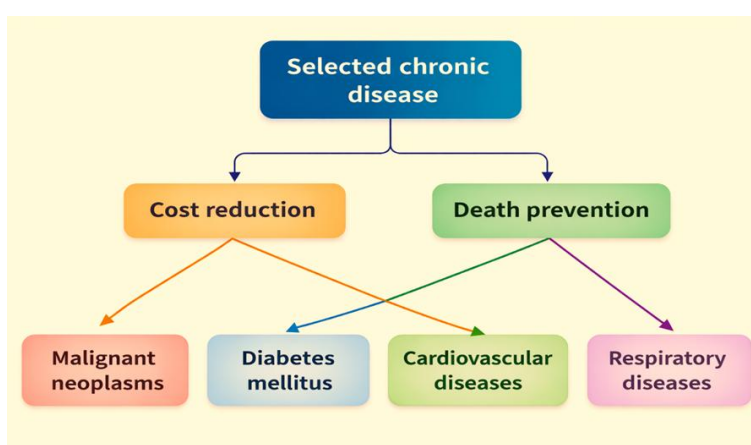


Fig.3 Fuzzy Analytic Hierarchy Process (FAHP) model for prioritizing chronic diseases based on IoT applicability.

5. Results and Discussion

This section presents the findings obtained from structural coding, Latent Dirichlet Allocation (LDA), and the Fuzzy Analytic Hierarchy Process (FAHP). The combined analysis provides a comprehensive understanding of IoMT technologies, research trends, and disease prioritization in chronic disease management.

5.1 IoMT Technologies for Chronic Disease Management

The analysis identified wearable sensors as the primary source of physiological data in IoMT systems. Devices such as continuous glucose monitors, smart insulin pumps, and fitness trackers enable continuous monitoring of glucose levels, heart rate, and physical activity.

Healthcare data are processed using cloud computing, big data analytics, and edge/fog computing. Cloud platforms provide scalable storage and advanced analytics, while edge and fog computing support low-latency processing and rapid clinical responses. Security is strengthened through blockchain technology for secure data sharing and semantic technologies that improve interoperability between healthcare systems. The overall architecture of the IoMT-based chronic disease management system is illustrated in Fig. 4.

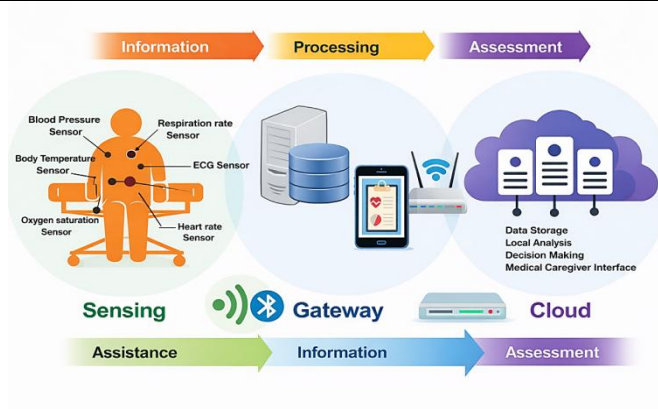


Fig. 4. Architecture of IoMT-based chronic disease management system.

5.2 Chronic Diseases Supported by IoMT

The literature shows that IoMT applications mainly focus on diabetes mellitus and cardiovascular diseases, reflecting their high global prevalence and the need for continuous monitoring. Recent studies also demonstrate growing interest in cancer, asthma, and mental health disorders, indicating the expansion of IoMT applications into broader healthcare domains. The disease ranking based on IoMT-enabled cost reduction is presented in Fig. 5.

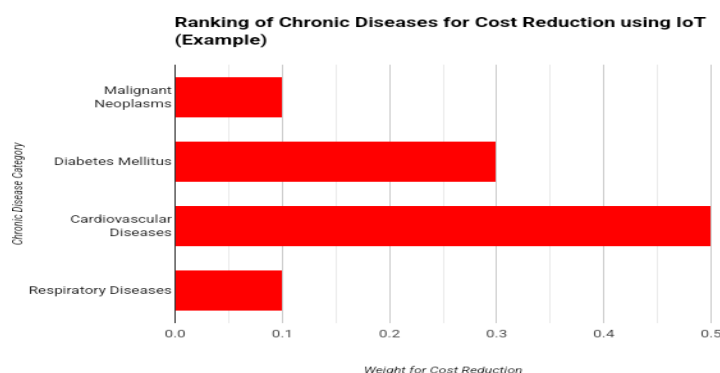


Fig.5 Ranking of Chronic Diseases for cost reduction using IoT

5.3 Topic Modelling Results

LDA identified approximately 18 major research topics, with patient monitoring, healthcare data management, sensors, and real-time monitoring emerging as the dominant themes. These findings indicate that current IoMT research is largely centred on data collection and processing. However, limited attention has been given to intelligent edge-based decision-making, highlighting an important research gap for future IoMT systems [19].

5.4 FAHP-Based Disease Prioritisation

FAHP ranked cardiovascular diseases as the highest priority for IoMT implementation, followed by diabetes mellitus. Cancer and respiratory diseases were ranked lower but remain important application areas. The consistency of the rankings across different evaluation criteria demonstrates the reliability of the FAHP model [25].

5.5 Discussion

Wearable sensing technologies remain the foundation of IoMT healthcare systems. Edge and fog computing are gradually replacing cloud-only architectures by enabling faster processing and reduced latency. For diabetes management, edge intelligence supports immediate detection of abnormal glucose levels, real-time alerts, and reduced dependence on cloud connectivity. Although data acquisition has been extensively studied, intelligent edge-based decision-making, TinyML integration, and interoperability remain important research challenges.

5.6 Future Research Directions

Future research should expand IoMT applications to diseases such as Parkinson's disease, Alzheimer's disease, epilepsy, and respiratory disorders. Further work is also required to develop edge-enabled IoMT systems, non-invasive biosensors, region-specific healthcare models, and stronger cybersecurity and privacy mechanisms to improve healthcare reliability and data protection.

6. Conclusion

This study presents a systematic analysis of the role of the Internet of Medical Things (IoMT) in chronic disease management using structural coding, Quantada-based text mining, Latent Dirichlet Allocation (LDA), and the Fuzzy Analytic Hierarchy Process (FAHP). The combined methodology provides a comprehensive understanding of current research trends, key technologies, and disease prioritization.

The findings show that wearable sensors, cloud computing, big data analytics, and edge/fog computing are the core technologies supporting IoMT-based healthcare systems. LDA identified approximately 18 major research themes, with patient monitoring and healthcare data management emerging as the most prominent areas.

The FAHP analysis identified cardiovascular diseases and diabetes mellitus as the highest-priority conditions for IoMT implementation because of their significant healthcare burden and the benefits of continuous monitoring. These results demonstrate the potential of IoMT to improve disease management through real-time monitoring, intelligent data analysis, and timely clinical decision-making.

Future research should focus on developing edge-enabled IoMT systems, non-invasive sensing technologies, and secure data management frameworks to improve healthcare efficiency and patient outcomes. Overall, this study provides useful insights for researchers, healthcare professionals, and policymakers in designing scalable and patient-centered IoMT solutions for chronic disease management.

References

- [1]. Zubiaga, A., Procter, R., & Maple, C. (2018). A longitudinal analysis of the public perception of the opportunities and challenges of the Internet of Things. *PLOS ONE*, 13(12), e0209472.
- [2]. Couturier, J., Sola, D., Borioli, G. S., & Raiciu, C. (2012). How can the Internet of Things help to overcome current healthcare challenges. *Digiworld Economic Journal*, 87, 67–81.
- [3]. Vermesan, O., Friess, P., Guillemin, P., Sundmaeker, H., Eisenhauer, M., Moessner, K., et al. (2014). *Internet of Things strategic research and innovation agenda*. IERC – European Research Cluster on the Internet of Things.

- [4]. Li, S., Xu, L. D., & Zhao, S. (2015). The Internet of Things: A survey. *Information Systems Frontiers*, 17(2), 243–259.
- [5]. Wang, S. H., Ding, Y., Zhao, W., Huang, Y. H., Perkins, R., Zou, W., et al. (2016). Text mining for identifying topics in the literatures about adolescent substance use and depression. *BMC Public Health*, 16, 279.
- [6]. Jagadeeswari, V., Subramaniaswamy, V., Logesh, R., & Vijayakumar, V. (2018). A study on medical Internet of Things and big data in personalized healthcare system. *Health Information Science and Systems*, 6(1), 14.
- [7]. Chou, D. (2016). What can IoT do for healthcare? Retrieved from <https://www.cio.com>
- [8]. Saldaña, J. (2015). *The coding manual for qualitative researchers*. Sage Publications.
- [9]. Haynes, E., Garside, R., Green, J., Kelly, M. P., Thomas, J., & Guell, C. (2019). Semi-automated text analytics for qualitative data synthesis. *Research Synthesis Methods*, 10(3), 452–464.
- [10]. de Moraes Barroca Filho, I., & de Aquino Junior, G. S. (2017). IoT-based healthcare applications: A review. In O. Gervasi et al. (Eds.), *Computational science and its applications – ICCSA 2017* (pp. 47–62). Springer.
- [11]. Turcu, C. E., & Turcu, C. O. (2013). Internet of Things as key enabler for sustainable healthcare delivery. *Procedia - Social and Behavioral Sciences*, 73, 251–256.
- [12]. Yang, K., & Meho, L. I. (2006). Citation analysis: A comparison of Google Scholar, Scopus, and Web of Science. *Proceedings of the American Society for Information Science and Technology*, 43(1), 1–15.
- [13]. Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., et al. (2018). Quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, 3(30), 774.
- [14]. Islam, S. M., Kwak, D., Kabir, M. H., Hossain, M., & Kwak, K. (2015). The Internet of Things for health care: A comprehensive survey. *IEEE Access*, 3, 678–708.
- [15]. Sugimoto, C. R., Li, D., Russell, T. G., Finlay, S. C., & Ding, Y. (2011). The shifting sands of disciplinary development: Analyzing North American LIS dissertations using LDA. *Journal of the American Society for Information Science and Technology*, 62(1), 185–204.
- [16]. Murali, N & Murugan, S & Sivakumar, K & Vivekanandan, Manojkumar & Sushith, Mishmala & Sridharan, Manikandan. (2024). Smart Commodities Public Distribution System using IoT. *Salud, Ciencia y Tecnología - Serie de Conferencias*. 3. 624. 10.56294/sctconf2024624.
- [17]. Sherman, R. (2014). *Business intelligence guidebook: From data integration to analytics*. Newnes.
- [18]. Paez, D. G., Aparicio, F., de Buenaga, M., & Ascanio, J. R. (2014). Big data and IoT for chronic patients monitoring. In *Proceedings of Springer* (pp. 416–423).
- [19]. MacQueen, K. M., & Guest, G. (2008). An introduction to team-based qualitative research. In *Handbook for team-based qualitative research* (pp. 3–19). AltaMira Press.
- [20]. Saaty, T. L. (2001). Analytic hierarchy process. In *Encyclopedia of operations research and management science* (pp. 52–64). Springer.
- [21]. N. Murali, R. Rajesh, S. Sridharan, A. E. P. Mariadas, S. P. Murugan and S. Manikandan, "A GPS-based Face Attendance Register System using Android Applications stored in the Cloud," 2024 11th International Conference on Computing for Sustainable Global Development (INDIACom), New Delhi, India, 2024, pp. 17-22, doi: 10.23919/INDIACom61295.2024.10498447.

-
- [22]. Wu, Y., Liu, M., Zheng, W. J., Zhao, Z., & Xu, H. (2012). Ranking gene-drug relationships using LDA. In *Biocomputing 2012* (pp. 422–433). World Scientific.
 - [23]. Haley, D. F., Vo, L., Parker, K. A., Frew, P. M., Golin, C. E., Amola, O., et al. (2017). Qualitative methodological approach. In A. O’Leary & P. M. Frew (Eds.), *Poverty in the United States: Women’s voices* (pp. 9–23). Springer.
 - [24]. Kalanithi, K. & Samuel, G. & Malathi, M. & R. Venkatesan,. (2025). Development and implementation of a model predictive control system for a solar parabolic trough plant influenced by an advanced meteorological disturbance model. *Scientific Reports*. 15. 10.1038/s41598-025-20459-y.
 - [25]. Ithaya Priya, B., Venkatesan, R., Mohamed Harees, M., & Surya Prabha, R. (2026). The right to food as an enforceable entitlement: International norms, constitutional practice, and structural challenges. In *Globalization, food systems, and legal responses: Governance, justice, and sustainability in a changing world* (Vol. 4, Issue 1). International Multidisciplinary Book Series. <https://doi.org/10.69889/IMBS.978-81-685212-2-3>
 - [26]. Dadkhah, M., Lagzian, M., & Santoro, G. (2019). How can health professionals contribute to IoT knowledge? *VINE Journal of Information and Knowledge Management Systems*, 49(3), 299–340.
 - [27]. Zubiaga, A., Procter, R., & Maple, C. (2018). A longitudinal analysis of IoT perception. *PLOS ONE*, 13, e0209472.